NDSU AGRICULTURAL RISK POLICY CENTER

Crop Insurance Penetration in the United States, 2025

ARPC REPORT 2026–04

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September 20, 2026

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Summary

This report documents long-run patterns in market penetration of the U.S. Federal Crop Insurance Program (FCIP). Using administrative data from the Risk Management Agency (RMA), the analysis examines program uptake primarily through insured acreage, with national trends supplemented by insured liability and aggregate coverage levels. In general, FCIP penetration has increased substantially over the past five decades. However, the pace of growth has varied across measures and policy periods. Early expansion followed the introduction of premium subsidies and the establishment of a public-private delivery system. Later growth was shaped by adoption of revenue insurance, supplemental coverage endorsements, and index-based products. The long-run increase in program exposure reflects broadening participation across crops and regions. Disaggregated results reveal pronounced heterogeneity across commodities and regions. Field crops largely operate in mature, high penetration markets, while many specialty crops exhibit lower average penetration, greater volatility, and episodic participation. In contrast, forage, rangeland, and pasture insurance have exhibited sustained and structurally driven growth, largely reflecting the rapid adoption of rainfall index products, particularly in western states.

Keywords: Federal Crop Insurance Program, market penetration, insured acreage, Pasture Rangeland and Forage, specialty crops

Recommended Citation Format: Francis Tsiboe, Hongxi Zhao, Dylan Turner, Rwit Chakravorty, Walker Davis, and Sandro Steinbach (2026). *Crop Insurance Penetration in the United States, 2025*. ARPC Report 2026–04. Agricultural Risk Policy Center, North Dakota State University.



Key Insights

- **National penetration has risen substantially over five decades, but the pace has been uneven and tied to specific policy episodes.** All four indicators (share of selected commodity acreage insured, insured liability relative to crop production value, aggregate coverage levels, and insured acres as a share of total farm acres) sit at or near long-run highs. Inflection points line up with the 1980 Act (premium subsidies and the public-private delivery system), the 1994 Act (revenue insurance and Catastrophic Risk Protection), the 2000 Agricultural Risk Protection Act, and the 2014 Farm Bill's supplemental coverage endorsements.
- **Field crops behave as a mature, high-penetration market, but the group average masks three distinct regimes.** Aggregate penetration changes only modestly over the sample. Underneath, cotton and canola are saturated, rice, corn, sunflowers, and sugar beets are still expanding, and barley and wheat are contracting or unstable. A single mean conceals these crop-specific trajectories.
- **Specialty crops and forage operate on entirely different participation regimes.** Most fruits, nuts, and vegetables show low average penetration, large year-to-year fluctuations, and episodic participation, with only a handful (oranges, cranberries, dry peas) reaching saturation. Forage, rangeland, and pasture, by contrast, has grown rapidly and persistently, almost entirely through adoption of the Pasture, Rangeland, and Forage (PRF) Rainfall Index product, which by the most recent crop year had reached roughly 70 percent of all PRF-eligible acres nationwide.
- **State-level growth has been geographically concentrated, with PRF adoption accounting for most of the regional pattern.** Nearly all states recorded positive average annual growth in penetration between 2007 and 2022, with the most rapid increases in the Mountain West and West South Central regions. For 2025, PRF adoption accounts for most of the observed gains in these states, illustrating how product design, not just subsidy levels, can shape the breadth and persistence of adoption.

Introduction

The Federal Crop Insurance Program (FCIP) is a central component of the U.S. agricultural risk management system. It provides subsidized protection against yield and revenue losses resulting from weather shocks, price volatility, and other production risks. Previous publications document the size of the FCIP (Turner et al., 2023; Skorbiansky et al., 2022; Glauber, 2013; Baldwin et al., 2026), which as of 2025 insured over 560 million acres representing approximately $200 billion in liability. Although the overall size of the FCIP is well established, a related and seldom asked question is how widely and intensively crop insurance is used.

Market penetration measures address this question by quantifying the share of agricultural land, or its value, enrolled in the program. Documenting penetration is important for three reasons. First, it distinguishes mature insurance markets, where participation is near saturation, from thin or emerging markets where adoption gaps remain. Second, penetration trends provide an institutional record of how policy reforms, subsidy adjustments, and product innovations translate into observable shifts in insurance use. Third, differences across crops and regions can reveal structural variation in production systems, risk exposure, and product suitability.

This report provides a descriptive assessment of FCIP market penetration in U.S. crop production. It summarizes national trends from 1975 to 2025, with attention to major legislative and administrative milestones, and then documents commodity and state level heterogeneity from 2006 to 2025. The report is descriptive and institutional in scope. It does not estimate causal effects or welfare impacts and instead establishes a consistent empirical baseline to inform future research, program monitoring, and policy discussion. Within this framework, the report addresses three core questions:

- How has FCIP market penetration evolved nationally over time?
- How does penetration vary across commodities, and which crops exhibit saturated, expanding, or volatile participation regimes?
- How does penetration vary across states, and what role do product innovations play in shaping regional adoption?

The findings show that at the national level, every penetration measure tracked since 1975 has risen, but

at different rates and in response to different policy episodes. The introduction of premium subsidies under the 1980 Act and the establishment of the public-private delivery system, the revenue-insurance and Catastrophic Risk Protection (CAT) provisions of the 1994 Act, the Agricultural Risk Protection Act of 2000, and the supplemental products introduced after the 2014 Act each align with distinct inflection points in the share of selected commodity acreage insured, in insured liability relative to crop production value, in aggregate coverage levels, and in insured acres as a share of total farm acres. By the most recent crop year in the data, all four indicators sit at or near their long-run highs.

Below the national trend, commodity-level results are highly heterogeneous. Field crops as a group behave as a mature market: average penetration remains high and changes only modestly over the sample, but the group contains saturated crops (cotton and canola), still-expanding crops (rice, corn, sunflowers, and sugar beets), and contracting or unstable crops (barley and wheat). Most fruits, nuts, and vegetables exhibit lower average penetration, larger year-to-year fluctuations, and episodic participation, with only a small number of crops (oranges, cranberries, dry peas) reaching saturated regimes.

At the state level, nearly all states recorded positive average annual growth in penetration between 2007 and 2022, with the most rapid increases concentrated in the West and broadly aligned with the spread of the Pasture, Rangeland, and Forage (PRF) Rainfall Index product. PRF is the clearest example of broad and sustained adoption in the FCIP. These results indicate that aggregate measures of program size mask meaningful differences in how intensively insurance is used across crops and regions.

Data and Methods

This report measures market penetration primarily as insured acreage relative to available agricultural land. Because acreage data are consistently available across crops, regions, and time periods, this measure serves as the basis for the commodity- and state-level analyses that follow. At the national level, the report also presents two supplementary indicators: insured liability relative to crop production value and aggregate coverage level. These indicators provide additional context on the program's financial scale and coverage intensity, but they are limited to the national analysis because reliable denominators for liability and coverage level are difficult to construct consistently for individual commodities and subnational units. Appendix A provides an extensive discussion of the three primary penetration measures used in this report.

The analysis combines FCIP business data from the U.S. Department of Agriculture (USDA) Risk

Management Agency (RMA) Summary of Business with four sets of denominator data: the USDA National Agricultural Statistics Service (NASS) Census of Agriculture for 2007, 2012, 2017, and 2022; annual NASS crop area and production surveys valued using NASS price-received series; Farm Service Agency (FSA) planted-acreage records; and USDA Economic Research Service (ERS) Farm Income and Wealth Statistics for the value of U.S. crop production. All underlying data are current as of September 15, 2026. FCIP data cover commodity years 1975-2025 for national-level series and 2006-2025 for commodity- and state-level series.

The unit of analysis varies by section. National trends are measured by crop year. Commodity-level analysis uses commodity-by-crop-year observations, with commodities grouped into Field Crops, Fruits, Nuts, Vegetables, and Forages. State-level analysis uses state-by-census-year observations, with growth measured across the 2007-2012, 2012-2017, and 2017-2022 intercensal periods. The PRF analysis uses county-by-crop-year observations beginning in 2016.

Together, the primary penetration measure (insured acreage) and the two national-level supplementary indicators (liability share and aggregate coverage level) distinguish (i) the land footprint of insured activity, (ii) the financial exposure insured relative to national crop production value, and (iii) the coverage intensity among participants. This distinction matters for interpreting penetration trends: the FCIP can become "more penetrated" through deeper coverage choices and rising insured values even when insured acreage grows modestly, and policy changes may affect these margins differently.

National Penetration

The farm bills and policy acts in the past fifty years substantially expanded the safety net for producers (Zulauf et al., 2024; Mark et al., 2020; Bekkerman et al., 2019). Figure 1 presents a national overview of FCIP market penetration from 1975 to 2025. It presents multiple measures of market penetration and identifies major legislative milestones in crop insurance (see Table 1, Table 2, and Table 3), illustrating how successive policy reforms have shaped overall penetration patterns over time (Rosch, 2021).[1]

The institutional history of the FCIP frames these patterns. Federal crop insurance was first authorized in the Agricultural Adjustment Act of 1938 as a "bold experiment" in nationwide multiple-peril ("all-risk")

[1] Sugarcane is a perennial crop that may be harvested multiple times. For better representation of annual acreage, harvested acres are used instead of planted acres for this commodity.

coverage that no private company had successfully operated at scale, with the program initially restricted to wheat and a small set of commodities (Kramer, 1983; Sanderson, 1943). For its first 40 years, the program operated as a limited pilot: by 1980 only about half of U.S. counties and 26 crops were eligible for coverage, and insured liabilities and participation remained modest (Knight and Coble, 1997; Glauber, 2013). The aggregate context in Figure 1 is therefore one of a long, gradual pre-1980 baseline followed by successive expansions tied to specific legislative acts.

Table 1: ***Major Federal Crop Insurance Program Legislation, Establishment and Limited-Pilot Era.***

Year	Action	Major Changes to FCIP
1938	Agricultural Adjustment Act of 1938	Established the Federal Crop Insurance Corporation (FCIC). Authorized insurance for wheat.
1941	55 Stat. 255	Authorized insurance for cotton.
1944	Agricultural Appropriation Act of 1944	Prohibited FCIC from issuing any policies.
1944	58 Stat. 918	Reauthorized FCIC to issue policies for wheat, cotton, and flax. Established an experimental program to insure other major commodities in geographically limited areas.
1947	61 Stat. 718	Restricted FCIC policies for all crops to limited areas to improve actuarial performance.

Note: Table summarizes early statutory changes that established FCIC and limited crop insurance availability before the modern expansion period.

Source: NDSU Agricultural Risk Policy Center summary based on Congressional Research Service Report R46686 and public law records.

In the decades following the Great Depression and the Dust Bowl, the FCIP developed gradually, with participation largely concentrated among major field crops in core production regions. From 1975 to 1980, the share of selected commodity acreage insured grew on average by 2.08 percent per year. The Federal Crop Insurance Act of 1980 marked the start of an initial expansion phase between 1981 and 1993, introducing a 30 percent premium subsidy at the 65 percent coverage level and establishing a public-private delivery system with Approved Insurance Providers (AIPs) (Kramer, 1983). The 1980 Act was explicitly intended to make crop insurance the primary form of catastrophic protection available to producers, replacing the standing disaster assistance programs that had operated for selected crops since

1973 (Glauber, 2013; Glauber, 2004). Following the 1980 Act, premium subsidies and the public-private delivery system coincided with broader producer participation, although Congress had originally envisioned a 50 percent participation rate by the end of the decade and actual participation grew far more slowly than that target (Glauber, 2013; Glauber, 2004). The share of selected commodity acreage insured (see the green line in Figure 1) rose from 8.68 percent in 1980 to 30.96 percent in 1993. Liability as a share of production value increased from 4.68 percent to 13.74 percent, and insured acres as a share of total farm acres grew from 2.53 percent to 8.57 percent. Across these metrics, annual penetration increased on average by approximately 10.71-13.53 percent per year.

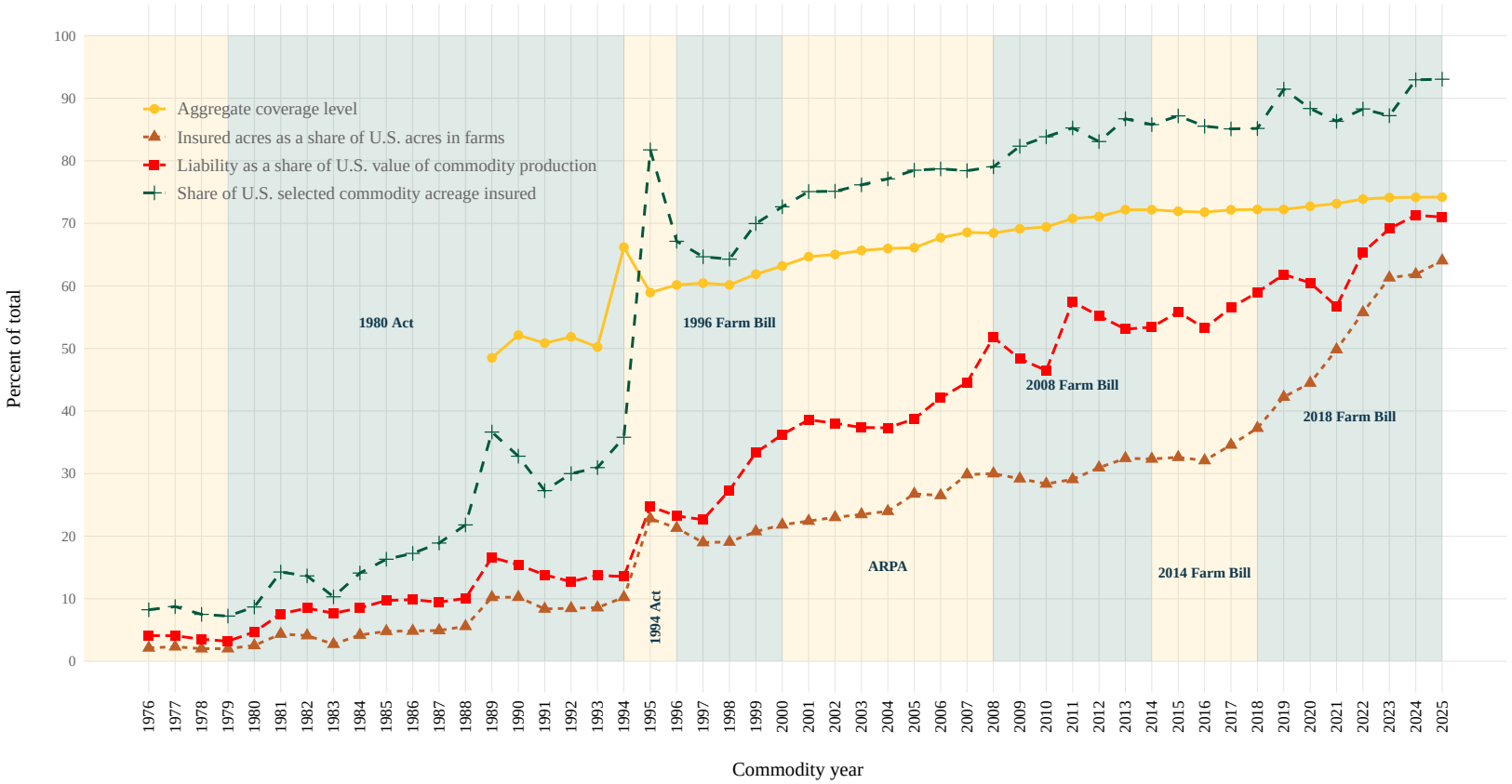
Table 2: ***Major Federal Crop Insurance Program Legislation, Subsidy Introduction and Modern Reform.***

Year	Action	Major Changes to FCIP
1980	Federal Crop Insurance Act of 1980	Authorized FCIC to issue policies for all commodities and geographic areas. Subsidized 30 percent of insurance premiums. Allowed private companies to sell FCIC policies.
1985	Food Security Act of 1985	Introduced conservation compliance requirements for crop insurance purchases.
1994	Federal Crop Insurance Reform and Department of Agriculture Reorganization Act	Increased premium subsidies and introduced catastrophic coverage. Required crop insurance purchases for eligibility for commodity support. Authorized prevented planting coverage.
1996	Federal Agriculture Improvement and Reform Act of 1996	Removed requirement linking commodity support and conservation compliance to crop insurance purchases. Created USDA's Risk Management Agency. Disallowed sales of crop insurance through USDA county offices unless approved insurance provider coverage is unavailable.

Note: Table summarizes the transition from limited federal insurance availability to the modern subsidized public-private delivery system.

Source: NDSU Agricultural Risk Policy Center summary based on Congressional Research Service Report R46686 and public law records.

Figure 1: *Nationwide Penetration of the United States Federal Crop Insurance Program, 1975 to 2025.*



Note: Aggregate coverage level equals total FCIP insured liability divided by potential liability. Production value is reported by calendar year; FCIP outcomes are reported by crop year. Selected acreage includes major row crops and sugarcane. 1980 Act stands for Federal Crop Insurance Act of 1980 [P.L. 96-365], 1994 Act stands for Federal Crop Insurance Reform and Department of Agriculture Reorganization Act of 1994 [P.L. 103-354], 1996 Farm Bill for Federal Agriculture Improvement and Reform Act of 1996 [P.L. 104-127], ARPA for Agricultural Risk Protection Act of 2000 [P.L. 106-224], 2008 Farm Bill for Food, Conservation and Energy Act of 2008 [P.L. 110-246], 2014 Farm Bill for Agricultural Act of 2014 [P.L. 113-79], 2018 Farm Bill for Agricultural Improvement Act of 2018 [P.L. 115-334].

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from USDA Risk Management Agency, USDA National Agricultural Statistics Service, and USDA ERS Farm Income and Wealth Statistics, as of September 15, 2026.

Between 1994 and 2000, FCIP participation continued to expand, particularly after the Federal Crop Insurance Reform and Department of Agriculture Reorganization Act of 1994. By the end of the 1980s policymakers concluded that the subsidy levels provided under the 1980 Act were not sufficient to achieve the 50 percent participation target without either making purchases compulsory or substantially increasing the subsidy rate (Glauber, 2013). Widespread crop losses from the 1993 Midwest floods, following an earlier round of ad hoc disaster legislation in 1988-1989, prompted the 1994 reform (Glauber, 2004). The 1994 Act increased premium subsidies at higher coverage levels, introduced revenue insurance products, and established CAT coverage at 50 percent of approved yield and 60 percent of expected market price, with the CAT premium fully subsidized and a flat administrative fee charged per crop per county (Glauber, 2013). Although CAT coverage was initially required for eligibility in other farm programs, the Federal Agriculture Improvement and Reform Act of 1996 repealed this requirement and created the USDA RMA (Orden and Zulauf, 2015). Following these institutional changes, the share of selected commodity acreage insured rose sharply from 35.8 percent in 1994 to 81.75 percent in 1995, when more than 220 million acres—over 80 percent of eligible acres (excluding hay)—were enrolled (Glauber, 2013). The sharp increase was likely tied to the mandatory nature of FCIP participation for eligibility in other farm program benefits. The subsequent removal of this requirement coincided with a partial pullback among producers who were not interested in FCIP participation alone. Because CAT coverage was fully subsidized, however, it provided producers with a very low-cost entry point into the program and marked an important turning point for continued enrollment in subsequent years. After this change, participation moderated but remained substantially above its pre-1994 level, with the share of selected commodity acreage insured declining to 72.65 percent by 2000. Over the full 1994-2000 period, liability as a share of production value increased from 13.54 percent to 36.24 percent, while insured acres as a share of farm acres rose from 10.23 percent to 21.82 percent. Aggregate coverage levels averaged about 61.57 percent over the period, with a one-year dip of 10.94 percent in 1995. These shifts imply an annual penetration increase of about 19.81-20.91 percent per year, depending on the metric.

The Agricultural Risk Protection Act of 2000 supported additional expansion by increasing premium subsidies and broadening the list of insurable crops (Glauber, 2013). Beginning in 2000, the RMA implemented a continuous rating formula that more closely aligned premium rates with individual indemnity risk, resulting in producer-specific pricing (Coble and Barnett, 2013; Tsiboe and Turner, 2023; Tsiboe et al., 2025b). Lower effective insurance costs coincided with growing demand for FCIP coverage, particularly for revenue products, which had been authorized as pilots for soybean and corn in Nebraska and Iowa in 1996 and were offered nationwide by 2003 (Glauber, 2013). The expansion of insurable crops made the program available to a broader set of producers, including more specialty crop growers.

From 2001 to 2007, the share of selected commodity acreage insured increased from 75.08 percent to 78.42 percent, and insured acres as a share of total farm acres rose from 22.42 percent to 29.84 percent. Aggregate coverage levels also increased from 64.66 percent to 68.56 percent. Across metrics, these trends indicate modest annual growth in FCIP penetration of roughly 0.73-5.01 percent per year. Evidence also links the expansion of risk-management tools to measurable producer outcomes, including productivity gains (Cornaggia, 2013).

The Food, Conservation, and Energy Act of 2008 further shaped the FCIP through renegotiation of the Standard Reinsurance Agreement, which governs the financial relationship between the Federal Crop Insurance Corporation (FCIC) and AIPs (Glauber, 2013). Key revisions included adjustments to Administrative and Operating expense rate calculations, reduced premium subsidies for area-based plans, and elimination of Premium Reduction Plans. The 2008 Farm Bill also introduced a standing disaster program, the Supplemental Revenue Assistance Payments (SURE) program, that based disaster aid on a farm's crop insurance liability and covered losses that effectively formed the farm's deductible on its crop insurance coverage (Glauber, 2013). These changes were followed by modest gains in penetration, with annual growth at roughly 1.07-1.9 percent, depending on the metric. By 2013, the share of selected commodity acreage insured reached 86.72 percent, aggregate coverage averaged 72.18 percent, liability as a share of crop production value rose to 53.09 percent, and insured acres as a share of total farm acres reached 32.45 percent.

Following enactment of the Agricultural Act of 2014, RMA introduced new insurance products, including Supplemental Coverage Option (SCO) and Stacked Income Protection Plan (STAX) for cotton producers beginning with the 2015 crop year. Both products covered "shallow losses" that would otherwise not be covered by the underlying yield or revenue policy (Glauber, 2013). Between 2014 and 2017, insured acres as a share of farm acres increased by 2.33 percent, while liability as a share of production value rose by 2.02 percent. In contrast, the share of selected commodity acreage insured and aggregate coverage levels declined annually by 0.26 percent and 0.01 percent, respectively. By 2017, these measures had settled at 85.1 percent, 72.16 percent, 56.52 percent, and 34.58 percent. By this point penetration was sufficient that, in contrast to the ad hoc disaster legislation of the 1988 and 1989 crop years, Congress did not pass ad hoc relief for the 2012 drought, a widely cited indication that the FCIP had become the program's primary mechanism for catastrophic protection (Smith et al., 2017; Barnaby and Russell, 2016).

Table 3: ***Major Federal Crop Insurance Program Legislation, Subsidy Maturation and Recent Adjustments.***

Year	Action	Major Changes to FCIP
2000	Agricultural Risk Protection Act of 2000	Increased premium subsidies to present levels. Granted authority for revenue coverage and insurance for livestock. Authorized creation of 508(h) pilot programs.
2008	Food, Conservation, and Energy Act of 2008	Introduced enterprise and whole farm units. Increased subsidies for area coverage plans. Authorized renegotiation of the Standard Reinsurance Agreement for the 2011 reinsurance year.
2014	Agricultural Act of 2014	Introduced area-based shallow loss coverage and yield exclusion options. Relinked conservation compliance requirements.
2018	Agricultural Improvement Act of 2018	Introduced multicounty enterprise units and the Enhanced Coverage Option. Expanded coverage for specialty crops, added hemp as an insurable commodity, enhanced Whole Farm Revenue Protection, and improved access for beginning, veteran, and socially disadvantaged producers.
2024	508(h) Modification	Raised the Enhanced Coverage Option (ECO) premium subsidy rate from 44 percent for revenue coverage and 51 percent for yield coverage to 65 percent.
2025	One Big Beautiful Bill Act (OBBBA)	Raised Supplemental Coverage Option (SCO) and Enhanced Coverage Option (ECO) premium subsidies to 80 percent for eligible policies. SCO's maximum coverage level is scheduled to increase from 86 percent to 90 percent beginning with the 2027 crop year.

Note: Table summarizes subsidy maturation, coverage expansion, and recent statutory and administrative changes to FCIP.

Source: NDSU Agricultural Risk Policy Center summary based on Congressional Research Service Report R46686, Risk Management Agency materials, and public law records.

The current policy environment, shaped by the Agricultural Improvement Act of 2018, further strengthened the FCIP. The 2018 legislation expanded coverage for specialty crops, added hemp as an insurable commodity, and improved access for underserved producers, including beginning, veteran, and socially disadvantaged farmers and ranchers. It introduced new supplemental policies such as the Enhanced Coverage Option (ECO) and encouraged conservation practices including cover cropping.

Enhancements to Whole Farm Revenue Protection were designed to better serve diversified operations and diversified specialty crop operations. Congress also directed FCIC to explore new products addressing catastrophic weather-related losses to reduce reliance on supplemental disaster assistance. From 2018 to 2025, the share of selected commodity acreage insured increased from 85.17 percent to 93.04 percent. Liability as a share of crop production value fluctuated between 58.98 percent and 71 percent. Insured acres as a share of farm acres rose to 64.04 percent in 2025. Aggregate coverage levels increased modestly from 72.22 percent to 74.2 percent. Subsequent administrative and statutory actions through 2025—including the 2024 RMA increase to the ECO premium subsidy rate (Risk Management Agency [RMA], 2024), the One Big Beautiful Bill Act enacted July 4, 2025 (Turner and Arita, 2025), and the implementing RMA Manager's Bulletin MGR-25-006 (Risk Management Agency [RMA], 2025)—further reshaped the producer-facing premium structure.

These national trends show that FCIP market penetration expanded in distinct policy-driven phases rather than along a smooth path. Major legislative changes increased affordability, broadened product availability, and strengthened the public-private delivery system, contributing to large gains in insured acreage, insured liability, and coverage intensity over time (Glauber, 2013; Smith et al., 2017). The same legislative record also indicates that the geographic and commodity distribution of these gains has been uneven, with Great Plains and Corn Belt producers receiving the bulk of net program benefits over much of the post-1980 period (Babcock, 2008). By 2025, the FCIP had become a much broader and more financially significant risk-management program than it was in the late 1970s, though growth rates varied across policy periods and across the penetration measures used in this report.

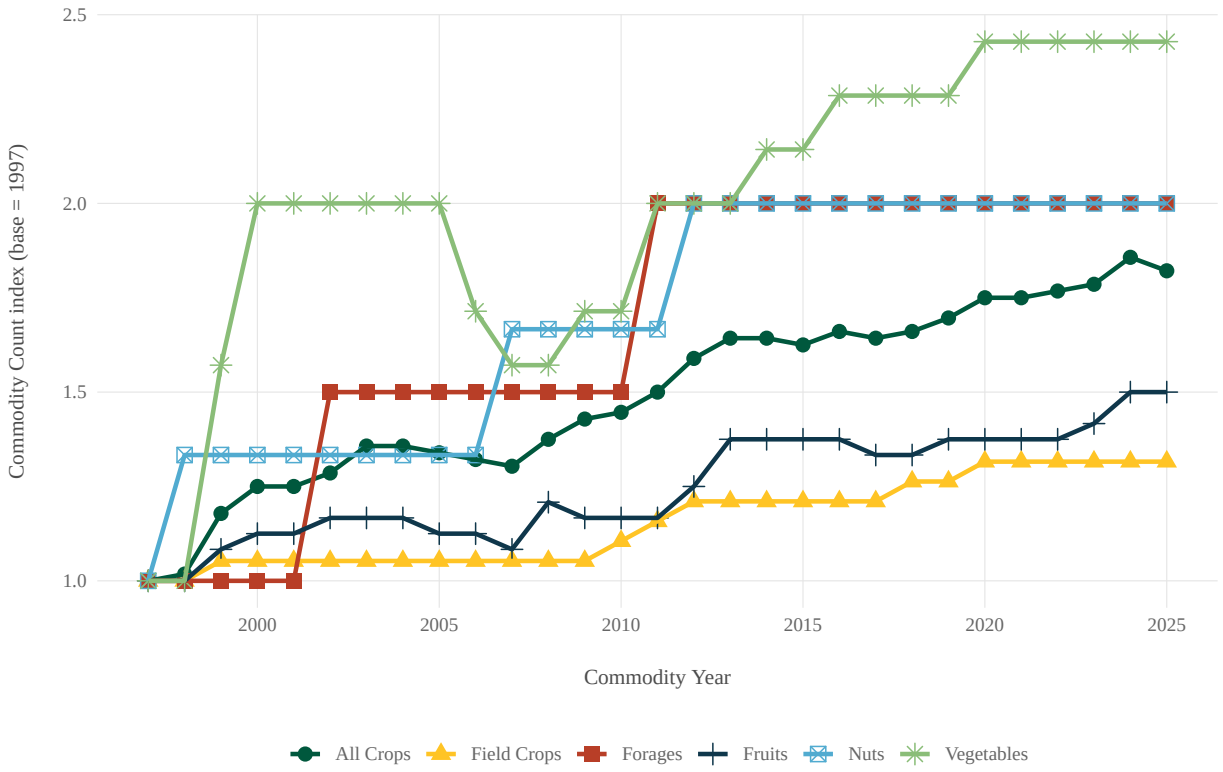
Commodity Acreage Penetration

As of 2025, the FCIP covered 102 different crops. To illustrate how commodity coverage has expanded over time, we construct a simple commodity count index. The index is normalized to 1997, such that values above one indicate that more crops are insured relative to 1997. Figure 2 demonstrates that the expansion in covered commodities has been primarily correlated with additions among specialty crops rather than by further expansion of traditional field crops.

In total, the number of insured crops increased by 82 percent from a base of 56 in 1997, nearly doubling over the period. However, this growth has not been evenly shared across crop groups. Vegetables account for the largest expansion, increasing from 7 to 17 covered crops. Growth was particularly rapid in the late 1990s and again following 2014. Nuts also doubled early on, growing from 3 to 6 crops by 2012

before leveling off. Fruits expanded more gradually, from 24 to 36 crops, reflecting steady but incremental additions over time. Forages increased in discrete steps, growing from 2 to 4 crops by 2011 and remaining stable thereafter. In contrast, field crops experienced only modest growth, increasing from 19 to 25 crops.[2] This pattern underscores that field crops were already broadly available for insurance when the program matured.

Figure 2: ***Count Index of Crops with Federal Crop Insurance Program Availability, Base Year 1997.***



Note: Index counts distinct commodities with any FCIP enrollment by commodity year, scaled so that 1997 equals one. Values above one indicate more insured crops relative to 1997. Livestock and animal products, controlled environment, nursery, and farm-level categories are excluded to focus on acreage-based commodities.

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

Although this expansion documents the widening scope of commodities eligible for coverage, it does not indicate how intensively producers use insurance across crops. Some commodities exhibit high

[2] Group figures are reported at the year each group's expansion leveled off (for example, nuts through 2012 and forages through 2011) and cover only the five commodity groups highlighted here; they therefore need not sum to the all-crops totals, which count every insured crop across all years through 2025.

enrollment and deep coverage, while others are technically insurable but participate at low rates. For this reason, the remainder of this section shifts from commodity counts to market penetration. Specifically, we examine the share of acres insured for each crop between 2006 and 2025, revealing economically meaningful differences in how producers utilize crop insurance across commodity groups.

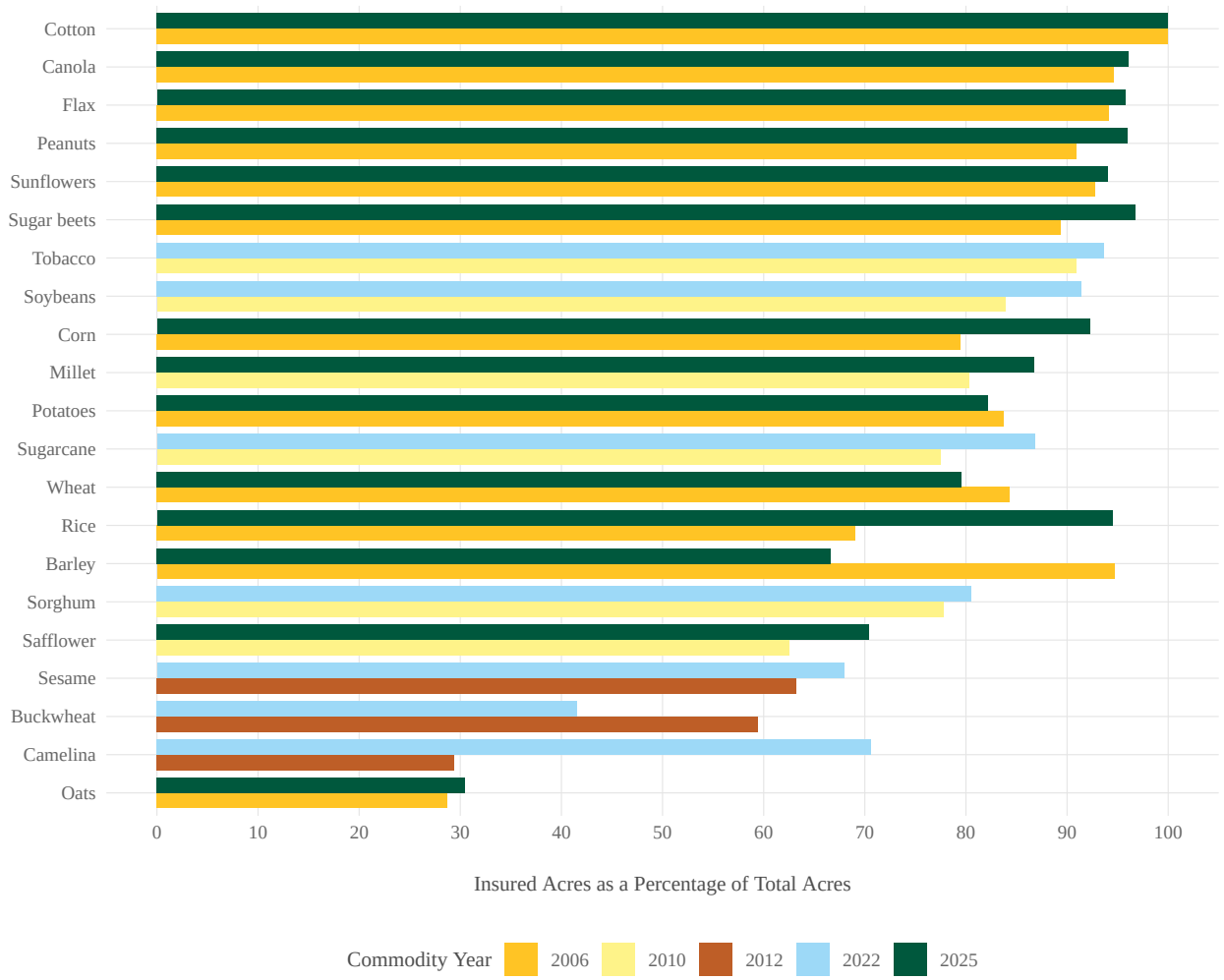
Penetration by Field Crops

Insurance penetration among field crops remains consistently high but varies across commodities in both level and trajectory between 2006 and 2025. As a group, field crops exhibit the characteristics of a mature insurance market. Average penetration, measured by insured acres as a percentage of total acres, averages 80.5 percent throughout the sample and changes only modestly over time, edging from 83.5 percent in 2006 to 84.4 percent in 2025. Although year-to-year fluctuations occur, the aggregate pattern reflects a stable participation regime rather than continued structural expansion. Figure 3 presents the heterogeneity of penetration within field crops. For each crop, the two bars indicate the earliest and most recent years, from 2006 through 2025, for which a penetration estimate can be computed—that is, years with sufficient acreage data to construct the denominator.

Within this group, cotton and canola evidently operate in a saturated regime, where the initial enrollment is already close to 100 percent. Cotton shows near universal participation throughout the period, with a mean penetration rate of 100 percent, a median of 100 percent, and very limited variability. Canola demonstrates similarly high and stable participation, averaging approximately 95.4 percent penetration, with only a one percentage point increase between 2006 and 2025. For both crops, insured acreage has little remaining scope for expansion.

Several major field crops, however, exhibit sustained growth in participation. Rice records the strongest expansion, with penetration increasing from 69 percent in 2006 to 94.5 percent in 2025, a cumulative increase of 25.5 percentage points. Corn also experiences large growth, rising from 79.4 percent to 92.3 percent, indicating continued deepening within an already high-penetration market. These trends suggest the popularity of insurance use among major row crops. Tobacco shows a meaningful increase from 90.9 percent to 93.6 percent. Flax shows a smaller gain, with penetration edging up from 94.2 percent to 95.7 percent.

Figure 3: ***Penetration of the United States Federal Crop Insurance Program by Field Crops.***



Note: Bars show insured acres as a share of available commodity acreage for the first and most recent commodity years in the 2006–2025 panel. Available acreage is based on an Olympic mean of NASS, Census, FSA, and related acreage sources, with RMA prevented-planting acres added. Values are capped at 100 percent.

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

Other crops display measurable gains despite greater short-run variation. Peanut penetration increases from 90.9 percent to 96 percent. Sunflowers reach near full participation by 2025. Sugar beets rise steadily from 89.4 percent to 96.7 percent, suggesting continued adoption even from relatively high initial levels. In contrast, several crops experience material declines or instability. Barley exhibits the most pronounced contraction, falling from 94.7 percent to 66.6 percent. Wheat declines more gradually, from 84.3 percent to 79.5 percent after peaking in the middle of the sample period.

Minor and emerging crops display higher volatility and intermittent participation. Millet shows large fluctuations and an overall upward trend. Oats remain persistently low in penetration throughout the period. For very small crops such as sesame, with 65.6 percent penetration, camelina at 50 percent, and buckwheat at 50.5 percent, large percentage changes reflect low initial baselines rather than evidence of mature market expansion.

Field crops fall into three broad participation regimes. The first includes saturated crops such as cotton and canola. The second includes expanding crops such as rice and corn. The third consists of contracting or unstable crops including barley and wheat. Distinguishing among these regimes is helpful for interpreting commodity-level participation patterns.

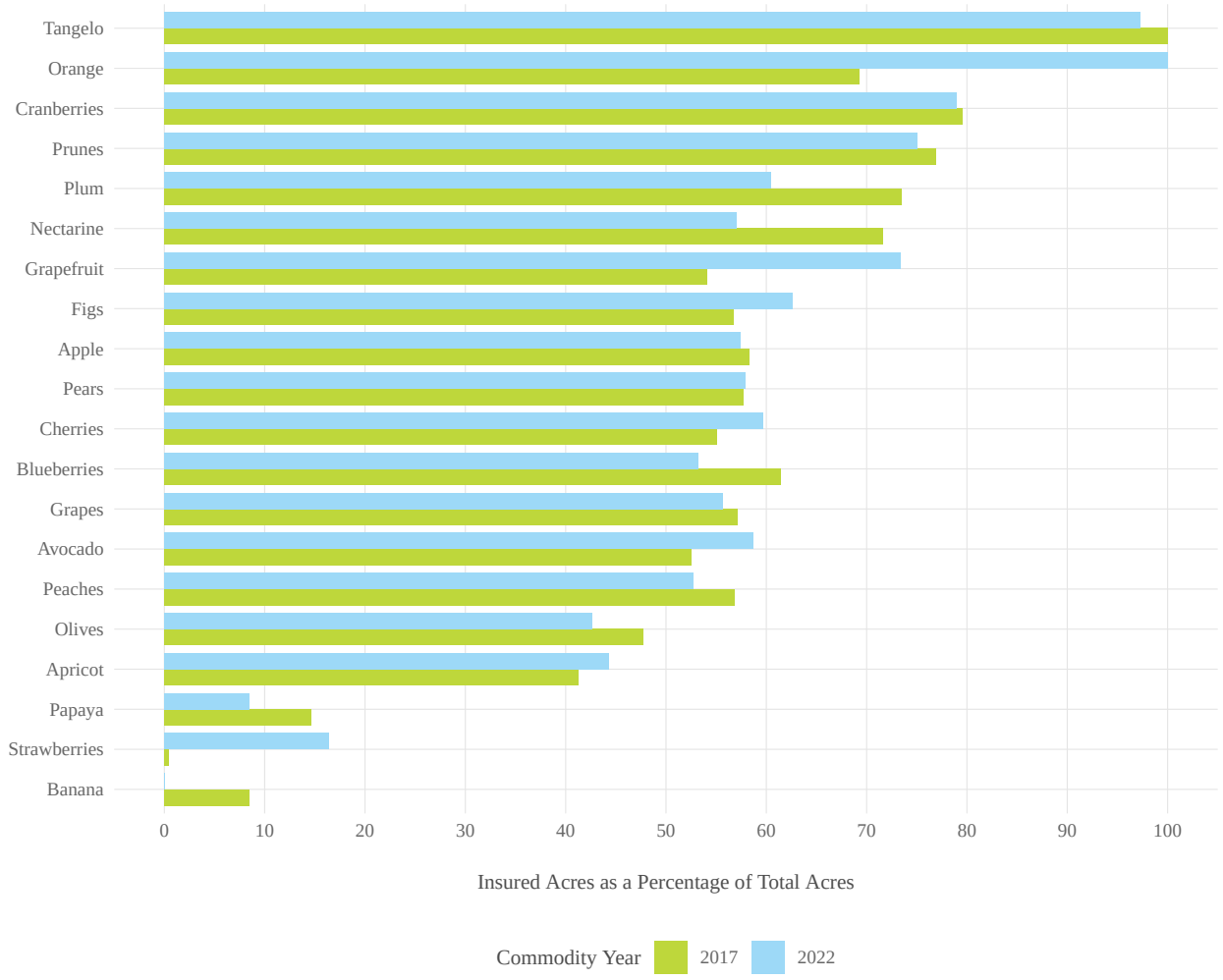
Penetration by Fruits

Insurance penetration among fruit crops varies considerably in both level and direction, although interpretation is constrained by limited reporting for many commodities. For most fruits, only two observations are available, typically for 2017 and 2022, which restricts the ability to draw strong conclusions about long-run trends. As a group, fruit crops display volatility rather than steady expansion. Average penetration increases from 54.7 percent in 2017 to 53 percent in 2022, but this rise reflects concentrated adoption in a small subset of crops rather than broad-based growth across the category. Accordingly, the observed increase should be interpreted as episodic uptake rather than evidence of systematic market maturation. Figure 4 presents market penetration using acreage for individual fruit crops, where only the first and last years for which a penetration estimate is available are plotted to improve figure readability.

A small number of fruit crops exhibit consistently high insurance participation. Oranges reach full penetration by 2022, rising from 69.2 percent in 2017. Cranberries maintain persistently high penetration of roughly 79.3 percent with little variation, while prunes show a slight decline from 76.9 percent to 75.1 percent. These crops operate near a saturated participation regime.

Several fruits experience notable gains over the observed period. Grapefruit increases from 54.1 percent to 73.4 percent, representing a considerable improvement in participation. Figs rise from 56.8 percent to 62.6 percent, and cherries increase from 55 percent to 59.7 percent, both indicating meaningful but moderate growth.

Figure 4: ***Penetration of the United States Federal Crop Insurance Program by Fruits, NASS Census Benchmark Years 2017 and 2022.***



Note: Bars show insured acres as a share of available commodity acreage for NASS Census benchmark years 2017 and 2022. Available acreage is based on an Olympic mean of NASS, Census, FSA, and related acreage sources, with RMA prevented-planting acres added. Values are capped at 100 percent. For fruits, most commodities are observed only in NASS Census years 2017 and 2022; the two bars therefore correspond to these benchmark years.

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

A broader group of fruits clusters around mid-range penetration levels with relatively modest movement. Apples decline slightly from 58.3 percent to 57.4 percent, pears remain essentially stable at 57.8 percent, and grapes fall from 57.1 percent to 55.7 percent. These changes are nontrivial in magnitude but do not represent large shifts in participation regimes.

More pronounced declines appear in several crops. Blueberries fall from 61.4 percent to 53.2 percent, olives decline from 47.7 percent to 42.6 percent, and peaches decrease from 56.8 percent to 52.7 percent, indicating meaningful erosion in insurance use.

Finally, very low-penetration fruits exhibit extreme percentage changes driven largely by small baselines and limited observations. Papaya declines from 14.7 percent to 8.5 percent. Strawberries display highly episodic participation, increasing from 0.4 percent to 16.4 percent in the most recent year, which is likely a low-base effect.

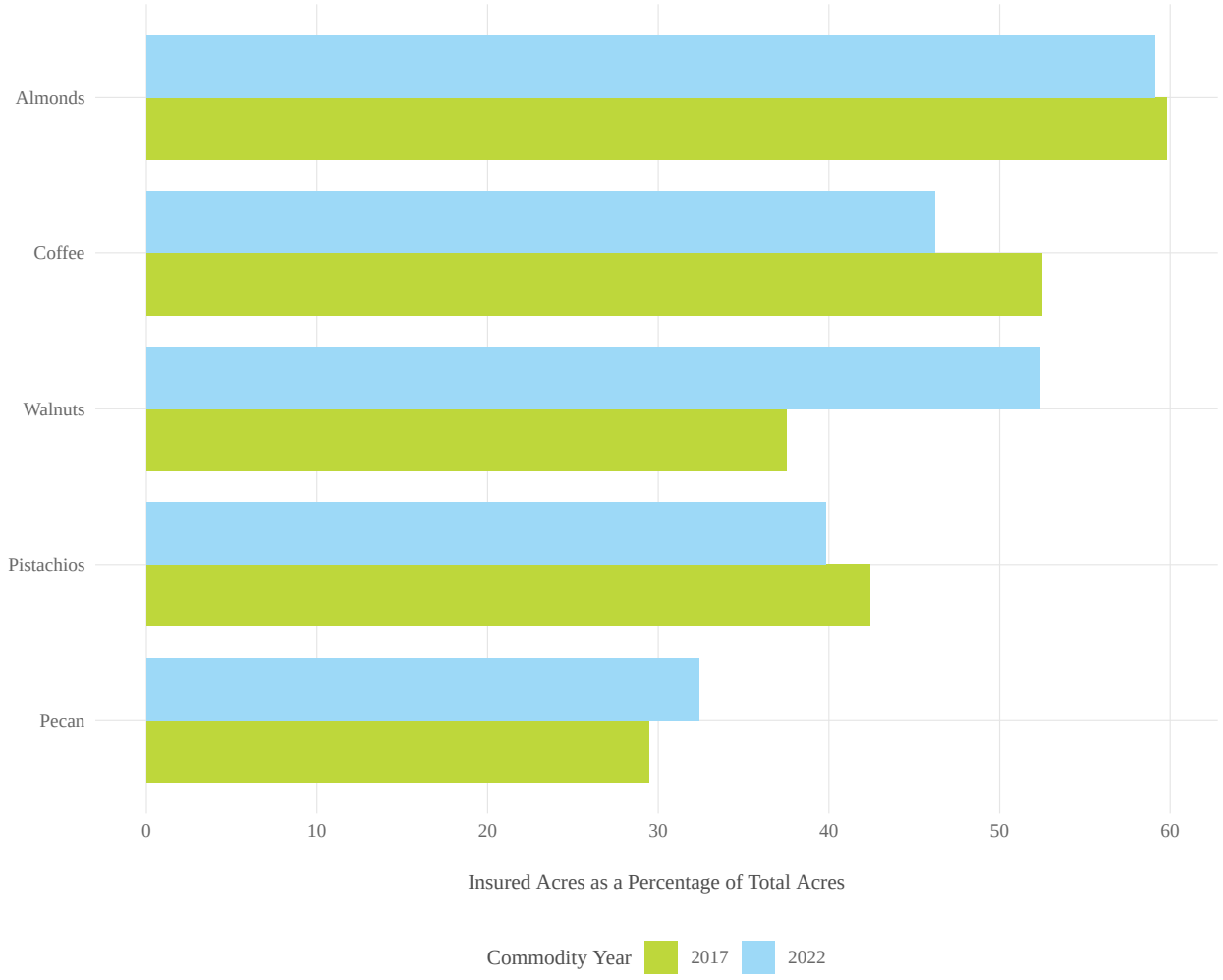
Fruit crops fall into three broad participation regimes. The first consists of high-penetration crops operating near saturation, such as oranges and cranberries. The second includes crops clustered around moderate penetration with limited movement, including apples, pears, cherries, and figs. The third comprises low-penetration crops characterized by episodic participation and high volatility, such as papaya and strawberries. Distinguishing among these regimes is important for understanding how insurance products perform across specialty crop markets and where adoption gaps are most evident.

Penetration by Nuts and Other Perennial Specialty Crops

As with fruit crops, interpretation of nut crop insurance penetration is limited by sparse data availability, as most nut commodities in the sample are observed only in 2017 and 2022. Within this constraint, overall insurance penetration among nut crops remains moderate and shows little evidence of sustained growth. Average participation is approximately 50.3 percent, with only a small net change (-1.8 percentage points) between 2017 and 2022. Although individual crops experience increases and declines during this period, these movements largely offset one another, indicating a relatively stable but unsaturated participation regime. Nevertheless, penetration levels and dynamics differ meaningfully across crops. Figure 5 presents market penetration using insured acres as a percentage of available agricultural land for individual crops.

Almonds exhibit stable mid-range penetration, with a mean of 59.5 percent and very low dispersion.

Figure 5: ***Penetration of the United States Federal Crop Insurance Program by Nuts and Other Perennial Specialty Crops, NASS Census Benchmark Years 2017 and 2022.***



Note: Bars show insured acres as a share of available commodity acreage for NASS Census benchmark years 2017 and 2022. Available acreage is based on an Olympic mean of NASS, Census, FSA, and related acreage sources, with RMA prevented-planting acres added. Values are capped at 100 percent. For nuts, most commodities are observed only in NASS Census years 2017 and 2022; the two bars therefore correspond to these benchmark years.

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

Pistachios experienced a modest decline, falling from 42.4 percent to 39.8 percent, indicating mild retrenchment over the observed window. In contrast, walnuts show a meaningful improvement in insurance use. Penetration increases from 37.5 percent to 52.4 percent, representing a sizable gain relative to the initial level. Pecans remain a low-penetration crop but record a meaningful increase from 29.4 percent to 32.4 percent, suggesting gradual uptake rather than stagnation.

Coffee is the only specialty crop in this category with more than two observed years and displays moderate penetration accompanied by pronounced volatility. Participation rises to a peak of 52.4 percent in 2017 before declining to 46.2 percent in 2022. While the net change over the period is small, the large intra-period swings point to episodic participation rather than steady expansion.

Nut crops fall into three participation patterns. The first consists of stable or mildly declining mid-range penetration, as observed for almonds and pistachios. The second includes crops with large growth from lower initial levels, such as walnuts. The third reflects moderate but volatile participation, as seen for coffee. Pecans remain in a low-penetration category but show gradual upward movement. These patterns highlight the importance of distinguishing structural participation levels from short-run fluctuations when evaluating insurance use among specialty crops.

Penetration by Vegetables

Vegetables are among the more rapidly expanding specialty-crop segments of the crop insurance program, although expansion begins from a relatively low baseline. Average insurance penetration increases from 34.8 percent to 37.6 percent over the observed benchmark years (2017 and 2022), indicating gradual broadening of participation across the group. However, this growth is uneven. Insurance use varies significantly across individual vegetables, and several crops exhibit sharp fluctuations alongside persistently low baseline coverage. Because many vegetable series are sparse and some penetration rates begin near zero, the analysis emphasizes economically meaningful changes rather than formal long-run trend estimation. Figure 6 presents market penetration using insured acres as a percentage of available agricultural land for individual vegetable crops.

At one extreme, dry peas operate in a fully saturated regime. Penetration remains at 100 percent in every

observed year, with no variation, indicating stable and entrenched insurance use.[3] Several vegetables show clear and meaningful expansion from low to moderate coverage. Cucumbers also show steady improvement, increasing from 22.6 percent to 27.6 percent, consistent with gradual strengthening of insurance use.

Other vegetables are characterized less by sustained growth and more by pronounced swings. Mustard reaches near universal coverage one year before declining, while sweet corn experiences early gains followed by a contraction. In both cases, year-to-year movements dominate the pattern, indicating episodic participation rather than stable adoption.

Lower-penetration vegetables remain clustered below 25 percent coverage. Cabbage shows modest improvement, whereas pumpkins and peppers remain lightly insured and volatile despite occasional increases. At the lower end of the distribution, sweet potatoes and dry beans exhibit very low participation and continued decline, underscoring persistent barriers to insurance use in certain vegetable markets.

Vegetable crops fall into three participation regimes. The first consists of saturated crops with universal coverage, such as dry peas. The second includes crops moving toward moderate penetration, such as cucumbers and cabbage. The third comprises a broad set of vegetables characterized by low coverage and high volatility. These distinctions are important for separating genuine expansion in insurance use from short-run fluctuations when evaluating specialty crop participation.

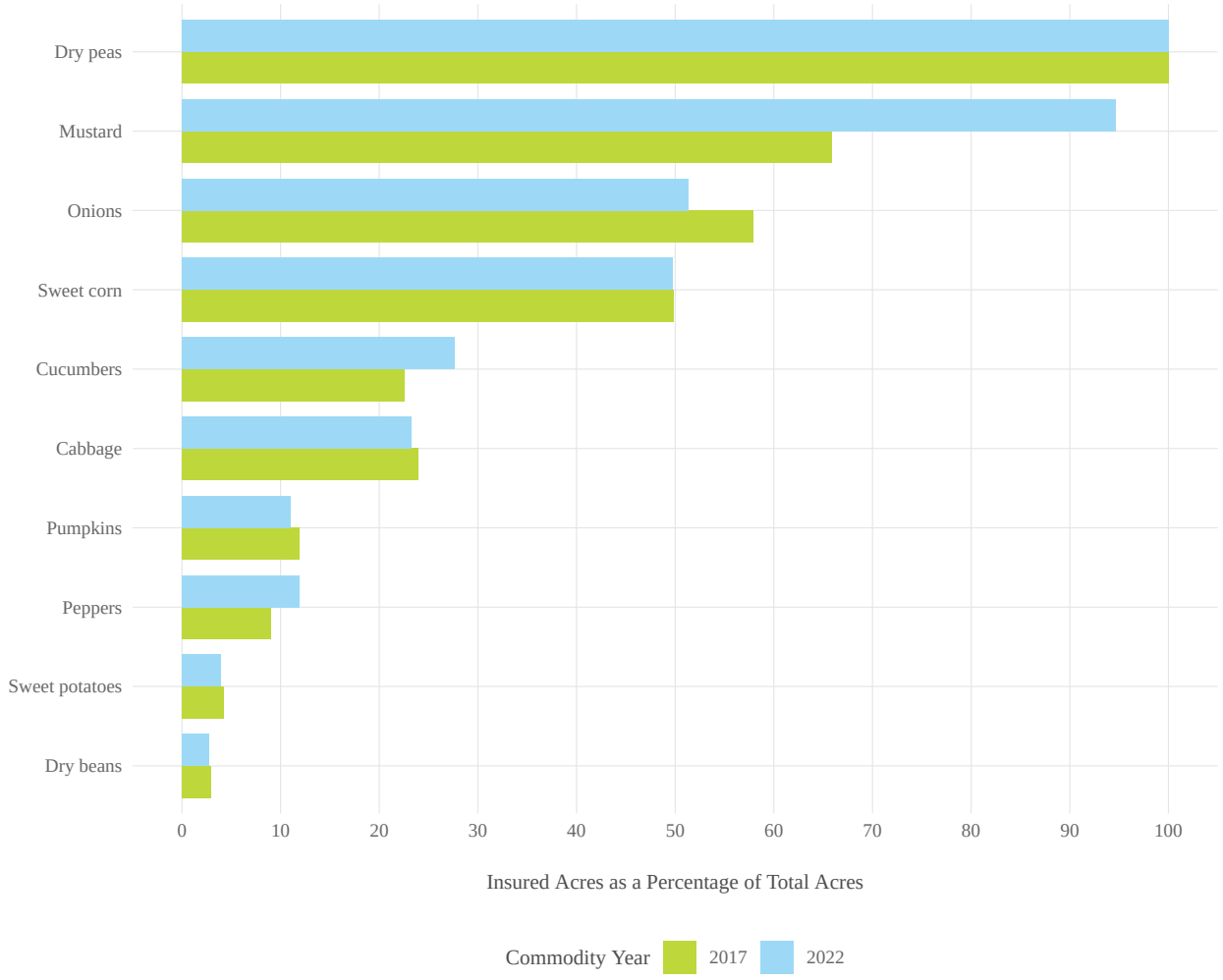
Penetration by Forage, Rangeland, and Pasture

Forage, rangeland, and pasture insurance represents the clearest example of broad and sustained expansion within the FCIP. Unlike field crops and many specialty crops, participation in this category rises steadily and persistently over time. Penetration increases from 6.5 percent in 2006 to 66.5 percent in 2025. Growth is gradual and continuous, with meaningful gains in nearly every year and no prolonged reversals. This pattern reflects structural adoption, with producers progressively incorporating insurance into their risk management strategies rather than responding to temporary incentives or short-term shocks.

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[3] Dry peas are normally categorized as pulses in agricultural works. This analysis adopts the USDA food classification, which puts dry peas under the vegetable subgroup of dry beans and peas.

Figure 6: ***Penetration of the United States Federal Crop Insurance Program by Vegetables, NASS Census Benchmark Years 2017 and 2022.***



Note: Bars show insured acres as a share of available commodity acreage for NASS Census benchmark years 2017 and 2022. Available acreage is based on an Olympic mean of NASS, Census, FSA, and related acreage sources, with RMA prevented-planting acres added. Values are capped at 100 percent. For vegetables, most commodities are observed only in NASS Census years 2017 and 2022; the two bars therefore correspond to these benchmark years.

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

The expansion in forage penetration is closely synchronized with PRF. PRF participation closely mirrors the overall forage trajectory, rising from about 10.19 percent of PRF-eligible acres in 2016 to 62.66 percent in 2025. Participation increases nearly every year, indicating widespread acceptance of index-based coverage as an accessible and practical risk management tool for forage producers. The magnitude and persistence of this increase suggest that PRF has evolved from a limited offering into a central component of forage insurance.

In contrast, forage coverage under Actual Production History (APH) policies remains limited. Although participation more than doubles in relative terms, it increases only from 0.54 percent in 2016 to about 1.22 percent in 2025 and remains below 2 percent throughout the period. Participation is uneven and characterized by occasional increases followed by declines, indicating that APH continues to play a minor role relative to PRF.

Index insurance may scale better for pasture, rangeland, and forage because individual forage production is difficult to observe and verify, especially when forage is grazed rather than harvested. APH coverage requires producer-specific yield histories and loss adjustment, while index insurance can use an external measure, such as rainfall relative to historical normal, to trigger payments. This lowers administrative costs and allows coverage to be offered more broadly.

Commodity-Level Penetration Regimes

Growth in the FCIP has occurred through two distinct dimensions: expansion in the number of covered commodities and variation in the extent to which those commodities are insured. Although the number of insurable crops has nearly doubled since the late 1990s, largely through the addition of specialty crops, insurance penetration remains highly uneven across commodities. Field crops generally operate within mature insurance markets characterized by high participation and limited room for further expansion. Nonetheless, meaningful differences persist among saturated crops, expanding crops, and those experiencing contraction. By contrast, most fruits, nuts, and vegetables exhibit lower average penetration, greater volatility, and more episodic participation. These patterns indicate that the availability of coverage does not automatically translate into sustained insurance use.

Forage, rangeland, and pasture insurance represents a distinct and contrasting regime. Participation increases steadily and at scale, aligned with the broad adoption of PRF. This experience illustrates the role of product design in shaping insurance uptake. Taken together, the results suggest that evaluation of the

FCIP benefits from considering not only the breadth of commodity eligibility but also the heterogeneity in participation regimes across crops. Expansion, saturation, and volatility reflect different insurance markets.

State-Level FCIP Penetration and Growth

This section examines crop insurance market penetration measured as insured acreage as a percentage of available agricultural land at the state level by aggregating insured acreage across all crops within each state. To ensure a comprehensive and consistent measure of total agricultural land, we use state-level agricultural acreage reported by the USDA Census of Agriculture for 2007, 2012, 2017, and 2022. For each state and census year, penetration is defined as the ratio of total insured acres across all crops to total agricultural land in that state. Figure 7 presents these results and highlights substantial heterogeneity in crop insurance penetration across states.

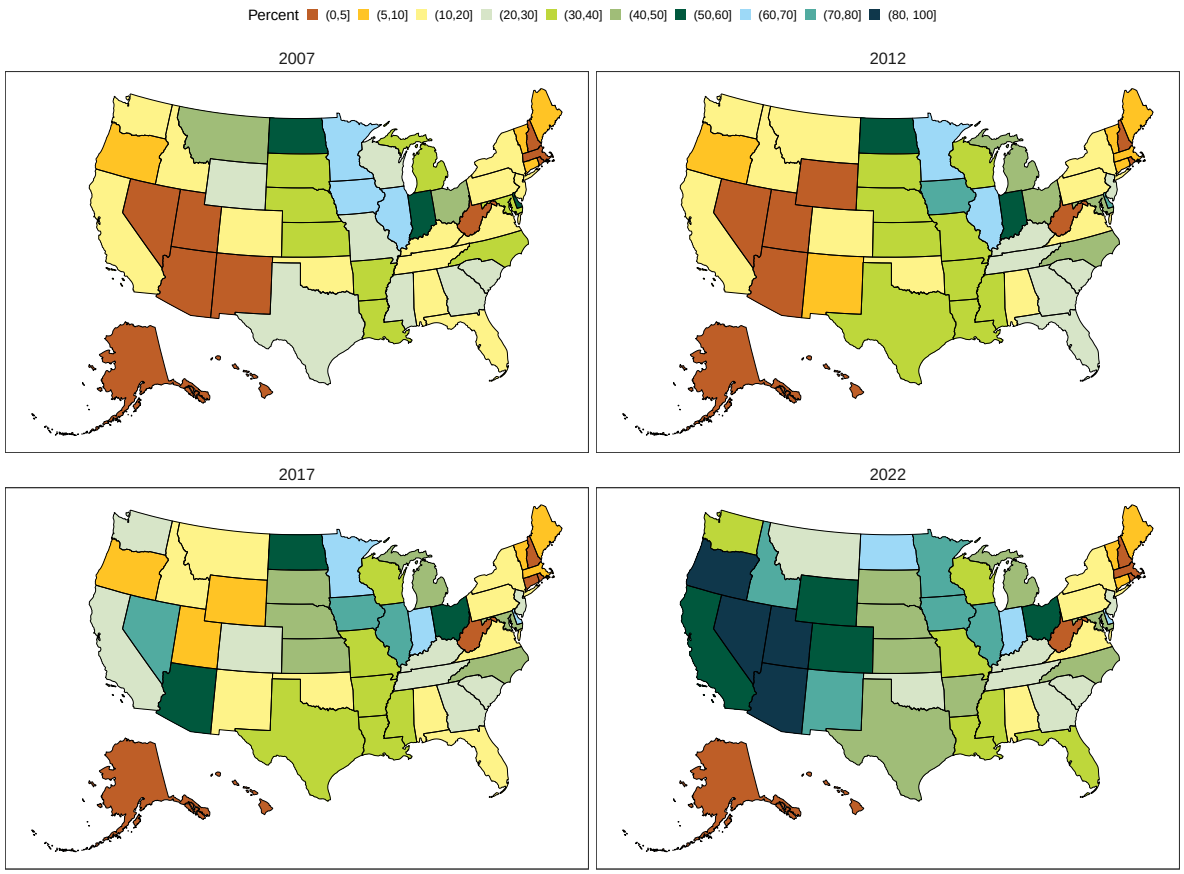
To characterize changes over time, we compute average annual growth rates in penetration from 2007 to 2022. These are defined as the mean of annual growth rates across the three intercensal periods: 2007-2012, 2012-2017, and 2017-2022. These growth rates are reported in Figure 8.

As of 2022, state-level FCIP penetration varies widely across the United States. Approximately 38.46 percent of states exhibit penetration rates of 50 percent or higher, indicating that at least half of agricultural land in those states is insured through the FCIP. States with the highest penetration include Utah and Oregon, each reaching nearly 100 percent penetration, followed closely by Idaho (77.5 percent). Other highly penetrated states include Illinois (76 percent), Minnesota (70.2 percent), North Dakota (66.3 percent), California (57.6 percent), and Ohio (53.8 percent).

A substantial share of states falls in the middle of the distribution, with insured acreage between 30 percent and 50 percent of total agricultural land. This group includes several major agricultural states such as Michigan (46.6 percent), Texas (41.5 percent), Arkansas (41.3 percent), Wisconsin (39.1 percent), Washington (37.6 percent), Mississippi (35.4 percent), and Louisiana (35 percent). These states demonstrate meaningful engagement with the FCIP but remain well below universal coverage.

At the lower end of the distribution, a small number of states report penetration rates below 5 percent, including Alaska (0.5 percent). Other low-penetration states reporting 5 to 20 percent acreage penetration include Virginia (16.8 percent), Pennsylvania (17.2 percent), and New York (17.9 percent). Low participation in these regions reflects structural differences in crop mix, farm structure, production

Figure 7: ***Insured Acres as a Percent of Total Agricultural Land by State, Census of Agriculture Years 2007, 2012, 2017, and 2022.***

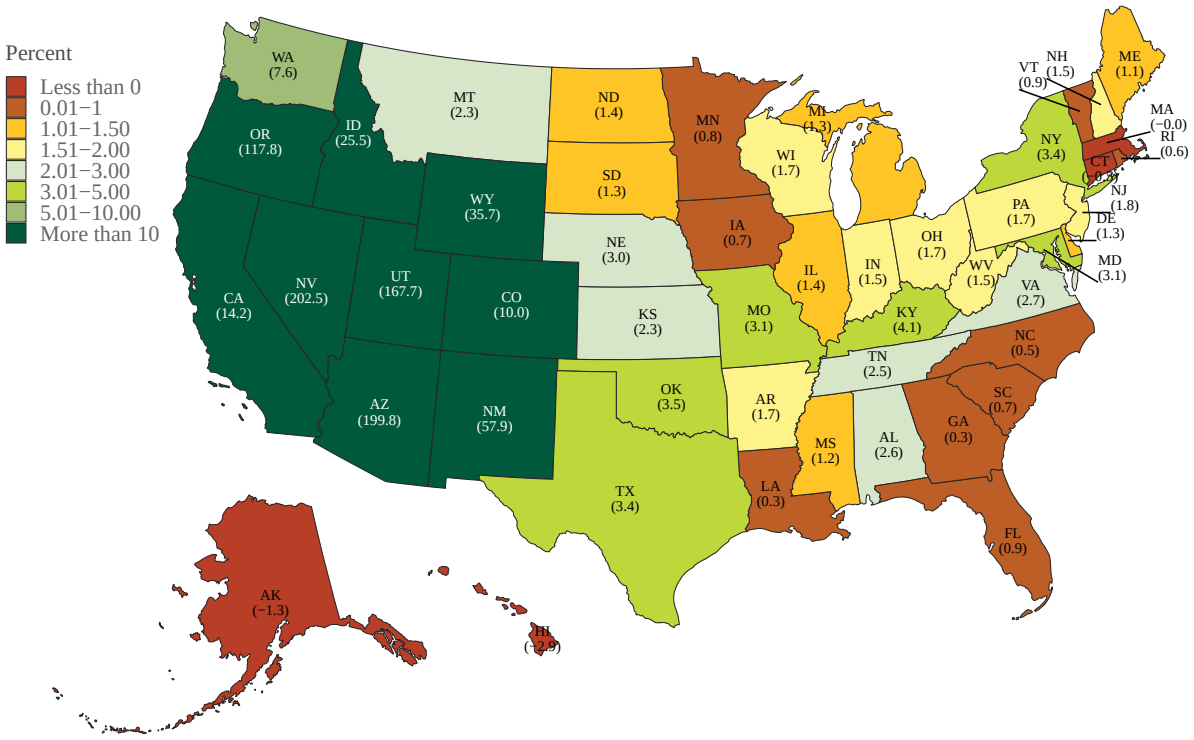


Note: Maps show insured acres as a percentage of total agricultural land by state for Census of Agriculture years 2007, 2012, 2017, and 2022. Insured acres include RMA-insured acres and prevented-planting acres for the corresponding crop year. Total agricultural land is from the USDA NASS Census of Agriculture. Values are capped at 100 percent. Alaska, Hawaii, and Puerto Rico are excluded from the map projection.

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

systems, and the relevance of available insurance products.

Figure 8: ***Annual Growth in Insured Acres as a Share of Total Agricultural Land by State, 2007 to 2022.***



Note: States are shaded by mean annual growth in FCIP penetration from 2007 to 2022. Growth is averaged across the 2007–2012, 2012–2017, and 2017–2022 intercensal periods and expressed in percent per year. Penetration is defined as insured acres plus prevented-planting acres as a share of total agricultural land. Parentheses report each state's mean annual growth rate.

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

Despite cross-state differences in levels, growth patterns over time are broadly positive. Figure 8 shows that nearly all states experienced positive average annual growth in penetration between 2007 and 2022, with especially rapid expansion concentrated in the western United States. States with average annual growth rates exceeding 10 percent include Nevada (202.5%), Arizona (199.8%), Utah (167.7%), Oregon (117.8%), New Mexico (57.9%), Wyoming (35.7%), Idaho (25.5%), California (14.2%), and Colorado (10%) (Tsiboe et al., 2025a). In most of these cases, rapid growth is correlated with sizable expansion in participation in PRF between 2017 and 2022. The expansion of PRF accounts for a substantial share of observed penetration growth in these states and reflects the increasing importance of index-based

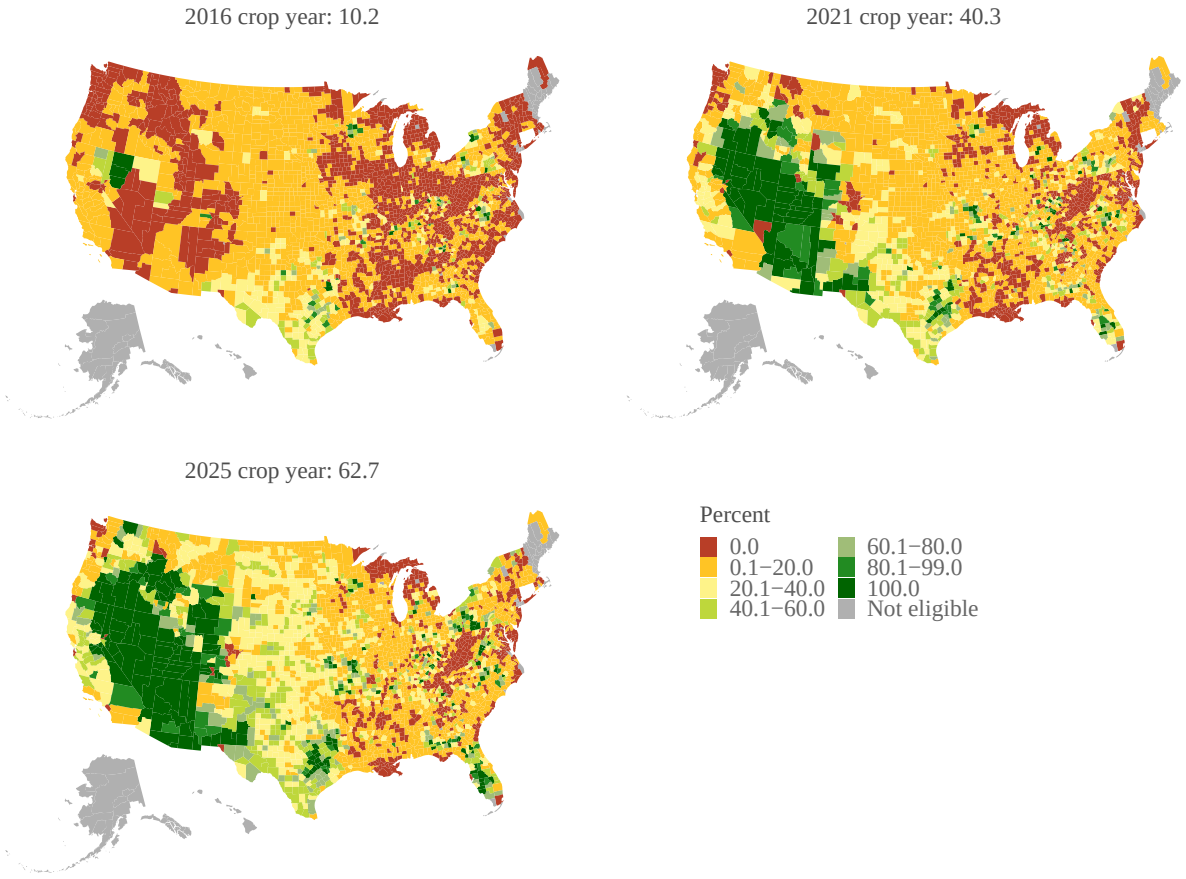
insurance products in rangeland-dominated production systems.

States with more limited rangeland exposure or lower PRF relevance exhibit more moderate growth trajectories. Several Midwestern and Southeastern states record average annual growth between 3 and 5 percent over the full sample period. By contrast, Alaska and Hawaii are the only states experiencing sustained declines in penetration between 2007 and 2022.

To further assess the role of PRF in shaping state-level penetration patterns, Figure 9 illustrates the evolution of PRF market penetration, measured as the share of PRF-eligible acres enrolled in PRF, across the continental United States for 2016, 2021, and 2025. Each map reports the share of insured acres enrolled in PRF relative to total PRF-eligible acres in each county. When PRF became available nationwide in 2016, participation was limited. Only about 11 percent of eligible acres were enrolled, and most counties in the eastern half of the country reported minimal or no participation. Early adoption was concentrated in the West South Central and West North Central regions, while the Mountain West initially exhibited relatively modest enrollment despite its strong forage base.

After 2016, PRF participation increased steadily, with national penetration rising from approximately 16 percent in 2017 to about 31 percent by 2019. This early expansion reflects growing awareness of the program and increased familiarity with rainfall index insurance as a forage risk management tool. Growth then accelerated through the early 2020s, with PRF penetration reaching about 44 percent by 2021 and roughly 63 percent by 2023 nationwide. Enrollment expanded rapidly across the Mountain West and West South-Central regions, concentrated in areas where forage production and livestock systems are most prevalent (Tsiboe et al., 2025a; Tsiboe and Davis, 2026). By 2025, nearly 70 percent of all PRF-eligible acres nationwide were enrolled, and many western counties reached full participation, indicating effective saturation.

Figure 9: ***Penetration of the Pasture, Rangeland, and Forage Program, Measured Against All PRF-Eligible Acres, 2016 to 2025.***



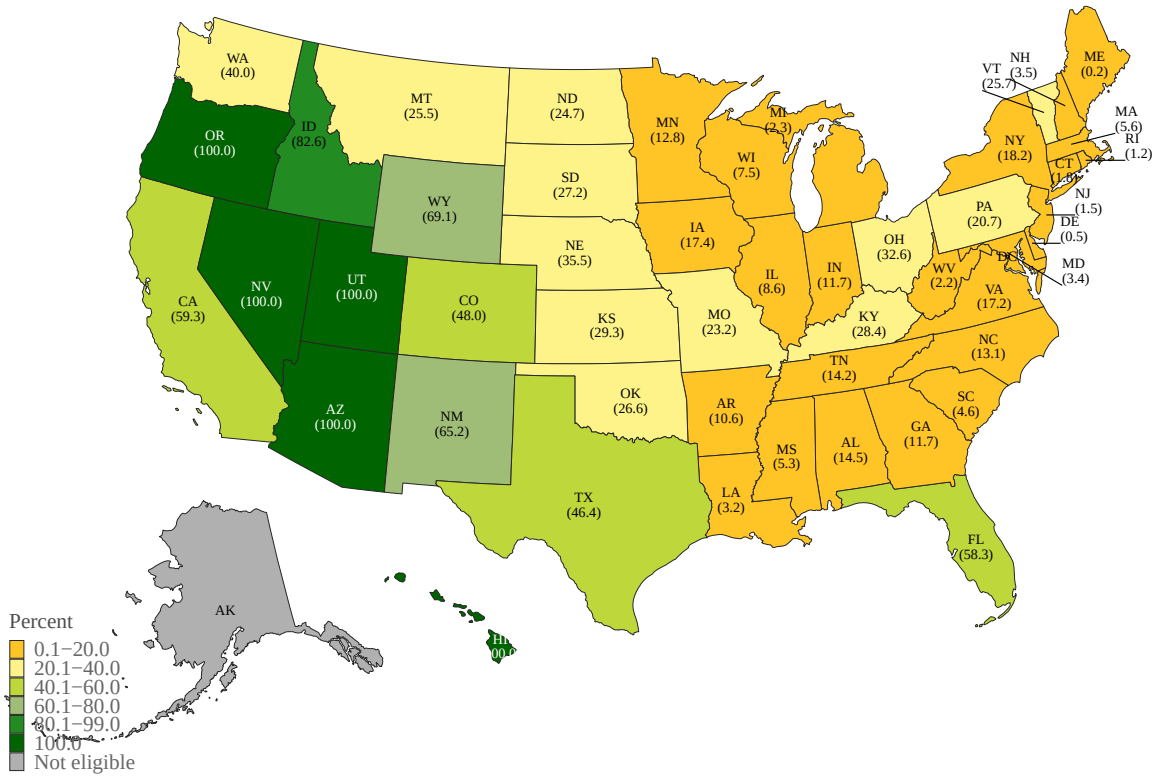
Note: County acreage eligibility for PRF is defined as the total land area within each PRF rainfall-index grid cell that satisfies specific land-cover criteria. Eligible land must contain either rangeland vegetation identified in the USDA Forest Service Rangelands V1 dataset or pasture, grass, shrub, or hay classes from the USDA NASS Cropland Data Layer (CDL; 2008–2024). In addition, every eligible pixel must meet an unpopulated-land requirement (population density equal to zero) based on the 2010 USGS block-level population-density raster, ensuring exclusion of developed and urbanized areas. To maintain consistency with grazing-use expectations, we assign ineligibility to grids in a county only if that county and all of its contiguous counties report zero January-1 inventory of cattle (including calves), goats, and sheep (including lambs) for all years from 2000 through the current year. Finally, preliminary county-level eligible acreage estimates produced by this procedure are capped at the maximum agricultural land acreage reported for each county in the Census of Agriculture (2002–2022).

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

Regions with the highest PRF enrollment correspond closely to areas dominated by forage production and limited alternative crop enterprises. In contrast, the Midwest, Southeast, and Northeast continue to exhibit low PRF penetration, with most states remaining below 25 percent enrollment, except for Florida,

Kentucky, and Ohio, as shown in Figure 10. These regions rely primarily on yield and revenue-based insurance products designed for row crop systems rather than rainfall index products targeted to forage.

Figure 10: ***2025 Penetration of the Pasture, Rangeland, and Forage Program, Measured Against All PRF-Eligible Acres.***



Note: County acreage eligibility for PRF is defined as the total land area within each PRF rainfall-index grid cell that satisfies specific land-cover criteria. Eligible land must contain either rangeland vegetation identified in the USDA Forest Service Rangelands V1 dataset or pasture, grass, shrub, or hay classes from the USDA NASS Cropland Data Layer (CDL; 2008–2024). In addition, every eligible pixel must meet an unpopulated-land requirement (population density equal to zero) based on the 2010 USGS block-level population-density raster, ensuring exclusion of developed and urbanized areas. To maintain consistency with grazing-use expectations, we assign ineligibility to grids in a county only if that county and all of its contiguous counties report zero January-1 inventory of cattle (including calves), goats, and sheep (including lambs) for all years from 2000 through the current year. Finally, preliminary county-level eligible acreage estimates produced by this procedure are capped at the maximum agricultural land acreage reported for each county in the Census of Agriculture (2002–2022).

Source: NDSU Agricultural Risk Policy Center (ARPC) using data from Risk Management Agency and National Agricultural Statistics Service, as of September 15, 2026.

Part of this regional contrast reflects the definition of PRF-eligible acres used in the analysis. An acre is classified as PRF-eligible if it was identified between 2008 and 2024 in the USDA Forest Service Rangelands dataset or classified as pasture, grass, shrub, or hay in the USDA NASS Cropland Data Layer.

Consequently, cropland intermittently used for forage or rotational grazing may be classified as eligible even if its primary use is row crop production. Low PRF participation in the eastern United States therefore likely reflects differences in production systems and suitability rather than lack of access.

The trajectory presented in Figure 9 and Figure 10 suggests that PRF adoption has moved from rapid expansion toward a slower phase of incremental growth. The stabilization of penetration rates and enrolled acreage since 2023 indicates that the program may be approaching saturation in its core pasture and rangeland regions. Continued growth in PRF, and in state-level FCIP penetration more broadly, may therefore depend less on geographic expansion in the West and more on participation in regions where enrollment has so far remained limited.

Conclusion

This report documents long-run patterns in FCIP market penetration using administrative data from 1975 to 2025. National penetration has increased substantially over this period, closely aligned with successive rounds of legislative reform that expanded premium subsidies, introduced revenue insurance, and broadened the set of insurable commodities. At the national level, insured acreage, liability relative to production value, and aggregate coverage levels have all risen, though at different rates and in response to different policy changes.

Beneath the national trend, commodity-level results vary considerably. Field crops generally represent a mature, high penetration market where the scope for further acreage expansion is limited. Within this group, however, some crops continue to gain while others, including barley and wheat, have experienced measurable declines. Fruits, nuts, and vegetables exhibit lower average penetration, greater year-to-year volatility, and more episodic participation. Insurance for pasture and forage represents a distinct case. Penetration in this segment has increased rapidly and persistently, largely synchronized with the introduction and adoption of PRF.

State-level results reinforce these patterns. The most rapid growth in penetration between 2007 and 2022 occurred in western states with large rangeland bases, where PRF adoption accounts for most observed gains. States with crop mixes primarily made up of row crops exhibit more moderate and stable penetration trajectories.

This report has several data limitations. Acreage-based penetration measures use total agricultural land as the denominator rather than strictly insurable acreage, which may understate participation relative to the

eligible land base. Liability is compared with total production value, but the two measures may not fully align in timing or commodity composition because production value may include commodities that are not insurable under FCIP. Data availability is uneven for some specialty crops, and the available data do not yet provide a detailed picture of the most recent crop year.

Several observations follow from these results. First, aggregate measures of program size, such as total insured acreage or total liability, can obscure meaningful variation in how intensively insurance is used across crops, regions, and producer types. Second, the distinction between saturated, expanding, and volatile participation patterns is relevant for program evaluation, since these categories reflect different insurance markets. Third, the experience of PRF illustrates that product design can shape the breadth and persistence of adoption even in segments where participation was historically limited. Future research could build on this descriptive baseline by examining the determinants of participation differences across commodities and regions, product suitability, the welfare implications of differences in coverage uptake across producer groups, and whether recent policy reforms have durably shifted participation patterns.

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Appendix A: Insurance Penetration Measures

(i) Insured acres as a share of available agricultural acres. This measure divides RMA-reported insured acreage by an acreage denominator that depends on the unit of analysis. At the national level the denominator is total U.S. land in farms from the USDA NASS farmlands series. At the state level the denominator is total agricultural land from the Census of Agriculture for the corresponding census year. At the commodity level the denominator is an "olympic mean" taken across the available crop-specific area sources within each commodity-year (NASS survey planted, harvested, and bearing acres; NASS track-record planted and harvested acres; NASS Census area harvested or bearing; FSA planted acres): when three or more sources are available the highest and lowest are dropped and the remaining sources averaged, so the trim is applied across the available sources within a year rather than across years. At the commodity level the insured-acreage numerator includes RMA prevented-planting acres, and the resulting share is capped at 100 percent. For pasture, rangeland, and forage the denominator is PRF-eligible acreage constructed county by county from the USDA Forest Service Rangelands V1 dataset and the USDA NASS Cropland Data Layer (pasture, grass, shrub, and hay classes), with a population-density screen and a contiguous-county livestock screen applied to exclude developed land and counties without grazing livestock. Because national and state-level denominators include pasture and other land not eligible for FCIP coverage, this measure should be read as a broad footprint measure rather than a conditional participation rate and will not mechanically reach 100 percent in practice.

(ii) Liability as a share of the value of U.S. crop production. This measure divides total insured liability by the total value of U.S. crop production. It captures the financial exposure insured through the program relative to national crop output. Differences between this measure and the acreage-based measure reflect uninsured production, deductibles, year-to-year price variation, and variation in insured value per acre. The limitation of this measure is that production value may include commodities not insurable under FCIP, and that ERS records production value on a calendar-year basis while RMA records crop-year totals; the two series therefore need not align exactly in timing.

(iii) Aggregate coverage level. This measure summarizes coverage intensity among enrolled participants. For each insured unit, potential liability is calculated as observed insured liability divided by the elected coverage level; total potential liability each year is the sum across units. The ratio of observed insured liability to potential liability is an aggregate measure of how close enrolled acres are, on average, to full coverage. A value of 75 percent means that, among insured exposure, elected coverage averages roughly three quarters of potential insurable liability. Units with missing or zero coverage-level values are dropped

before aggregation.

One additional derived measure is used in the report. State-level annual growth is defined as the mean of period growth rates across the three intercensal periods 2007-2012, 2012-2017, and 2017-2022, where each period growth rate is expressed as the percentage change in penetration divided by the number of years in the period.

The principal limitations of the analysis are as follows. First, the acreage denominators differ across sections (NASS farmlands, Census of Agriculture, olympic-mean crop area, and PRF-eligible acres), so penetration levels are not directly comparable across the national, state, commodity, and PRF panels. Second, several specialty-crop series are observed only in NASS Census years, limiting trend interpretation to the 2017 and 2022 benchmarks. Third, the national-level penetration measures use total land in farms as the acreage denominator and therefore include land not eligible for FCIP coverage, mechanically suppressing the level of the share.

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About the Agricultural Risk Policy Center

The Agricultural Risk Policy Center at North Dakota State University conducts independent, evidence-based economic research to inform agricultural policy and strengthen the farm safety net. Our work focuses on evaluating risk management tools such as crop insurance and disaster assistance, analyzing market disruptions, and providing timely insights that support agricultural producers, policymakers, and industry leaders.

Strengthening the U.S. Farm Safety Net with Evidence-based Policy Insights.

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